19A (+4C)
ELECTROCHEMISTRY

UNIT 03: WHAT REACTIONS PROVIDE

CHEMISTRY 1310

MIOY T. HUYNH

LEARNING OBJECTIVES

CHAPTER 19

1. Define and assign oxidation states or oxidation numbers.
2. Define aspects of redox reactions: oxidation, reduction, oxidizing agents, and reducing agents.
3. Write half-reactions from overall redox reactions, and vice versa.
4. Balance redox reactions.
5. Describe the components of voltaic (or Galvanic) cells using appropriate terminology and notation.
6. Calculate cell potentials using tables of standard reduction potentials.
7. Compare the strengths of oxidizing and reducing agents using tables of standard reduction potentials.
8. Describe the relationship between the standard cell potential and the equilibrium constant.
9. Perform calculations involving standard cell potentials and equilibrium constants.
10. Calculate the effects of concentration on cell potential using the Nernst equation.
11. Describe how electricity can be used to drive redox reactions.
12. Perform stoichiometric calculations involving mass and current in electrolytic applications.

Recall : Spontaneity, Driving Forces, and ΔG

Chemistry in Context

In Chapter 4, we said that reactions occur ***spontaneously*** due to a combination of changes in heat energy (ΔH_{rxn}) and randomness (ΔS_{rxn}).

We also noted that certain *patterns* of reactants and products tend to be associated with ***spontaneous reactions*** and classified these "patterns" into ***3 driving forces***.

Driving Forces	***Reaction Types***
1. Formation of a solid precipitate	→ Double-replacement reactions
2. Formation of water or neutralization	→ Acid-base reaction
3. Oxidation-reduction	→ Combustion, synthesis, decomposition, and single-replacements reactions

In Chapter 18, we learned $\boldsymbol{\Delta G_{\mathrm{rxn}}}$ **= "driving force"** and a ***spontaneous reaction*** is one in which $\boldsymbol{\Delta G_{\mathrm{rxn}} < 0}$.

Types of Chemical Reactions

4.2

TABLE 4.2 Summary of Reaction Types

Reaction Type	Generic Formula	Examples
Synthesis	$A + B \rightarrow AB$	$4\ Fe(s) + 3\ O_2(g) \rightarrow 2\ Fe_2O_3(s)$
Decomposition	$AB \rightarrow A + B$	$2\ H_2O(\ell) \rightarrow 2\ H_2(g) + O_2(g)$ $2\ KClO_3(s) \rightarrow 2\ KCl(s) + 3\ O_2(g)$
Single-Replacement	$A + BC \rightarrow AC + B$	$Zn(s) + 2\ HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$
Double-Replacement	$AB + CD \rightarrow AD + CB$	$2\ KI(aq) + Pb(NO_3)_2(aq) \rightarrow PbI_2(s) + 2\ KNO_3(aq)$ $HBr(aq) + KOH(aq) \rightarrow KBr(aq) + H_2O(\ell)$
Combustion	$C_xH_y + O_2 \rightarrow CO_2 + H_2O$	$C_3H_8(g) + 5\ O_2(g) \rightarrow 3\ CO_2(g) + 4\ H_2O(\ell\ or\ g)$

Oxidation States and Redox Reactions

4.6

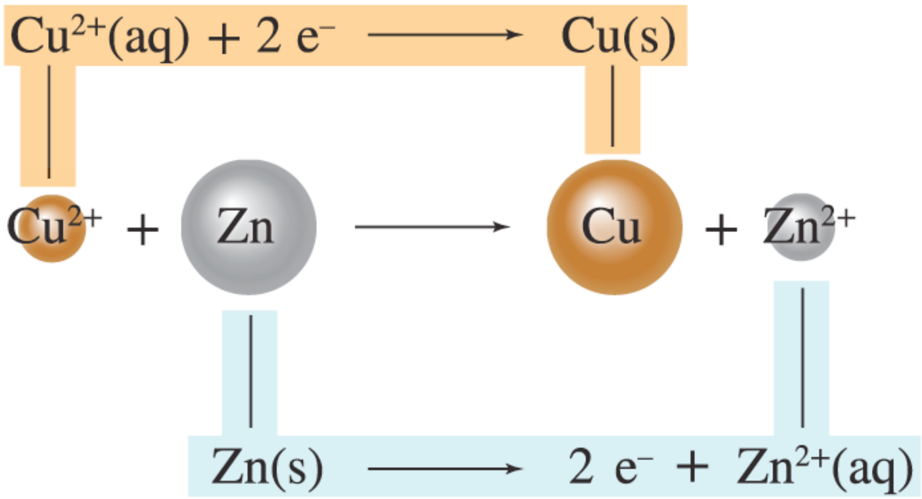
Reduction-oxidation reactions (or redox reactions) are those in which *electrons* (e^-) are transferred from one reactant to another reactant.

- Oxidation states (or oxidation numbers): A type of bookkeeping used to track the electrons transferred in redox reactions.

- Reduction

- Oxidation

FIGURE 4.24



Oxidation States and Redox Reactions

4.6

Basic (ordered) rules for assigning oxidation states:

0) Assume compound is 100 % ionic.

1) A neutral element that is *not* part of a compound has an oxidation state of _____.

2) The sum of the oxidation states in any compound/ion is equal to the overall charge on the compound/ion.

3) Monoatomic ions have oxidation states equal to their ionic charges.

4) In a compound, assign fluorine atoms an oxidation state of ____.

5) In a compound, assign hydrogen atoms an oxidation state of ____.

6) In a compound, assign oxygen atoms an oxidation state of ____.

Examples

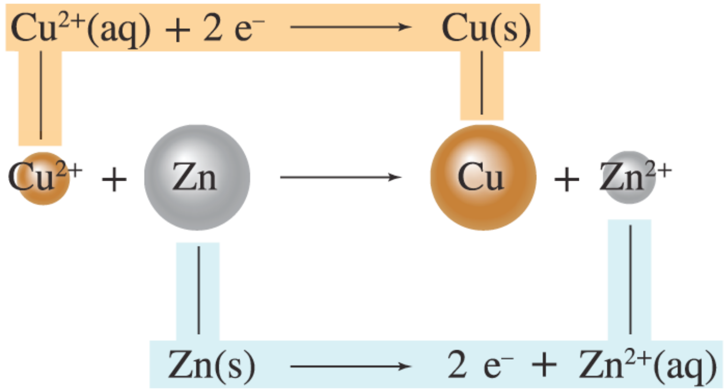
- Cu(s)
- F_2
- NaF
- NaClO
- HSO_3^-
- $Cr_2O_7^{2-}$
- $Fe(NO_3)_3$

Oxidation States and Redox Reactions

4.6

Reduction

- *Gain* of electrons (e^-)
- *Decreases* oxidation state
- Reactant species being reduced is called the **oxidizing agent** because it *causes oxidation.*



Oxidation

- *Loss* of electrons (e^-)
- *Increases* oxidation state
- Reactant species being oxidized is called the **reducing agent** because it *causes reduction.*

Redox Reactions

19.1

For simplicity/utility, **redox** reactions are *conventionally* split into **two half-reactions**.

- The **reduction half-reaction** shows the ***gain*** of electrons (e^-) by the species being reduced (i.e., the oxidizing agent).

- The **oxidation half-reaction** shows the ***loss*** of electrons (e^-) from the species being oxidized (i.e., the reducing agent).

Note: This convention is purely imaginative. Neither half-reaction can occur on its own.

Balancing Redox Reactions

19.2

Any two half-reactions can be added to give an overall balanced **redox** reaction.

- Use oxidation states to identify the reduction and oxidation half-reactions.
- Use "charge conservation" to determine the number of electrons (e^-) transferred.
- Balance the number of electrons: **the number of e^- gained in the reduction must equal the number of electrons lost in the oxidation.**

EXAMPLE PROBLEM

Learning Objective(s): 19.1, 19.2, 19.3, 19.4

Balance the redox reaction and identify the oxidizing and reducing agents.

$$Cl^{-} + Al^{3+} \rightarrow Cl_2 + Al$$

IN-CLASS QUESTION 1

Learning Objective(s): 19.1, 19.2, 19.3

For the balanced redox reactions below:

a) Identify the oxidizing agents and the reducing agents.

b) Determine the oxidation states of all atoms in all species.

c) Determine the total number of electrons transferred (n) in the redox reaction.

$$2\ Cr^{2+}(aq) + Co^{2+}(aq) \rightarrow 2\ Cr^{3+}(aq) + Co(s) \qquad n =$$

IN-CLASS QUESTION 2

Learning Objective(s): 19.1, 19.2, 19.3

For the unbalanced redox reaction below:

a) Identify the oxidizing agents and the reducing agents.

b) Determine the oxidation states of all atoms in all species.

c) Determine the total number of electrons transferred (n) in the redox reaction.

$$Mg(s) + FeSO_4(aq) \rightarrow MgCl_2(aq) + Fe(s) \qquad n =$$

ΔG … Again

Chemistry in Context

The change in Gibbs free energy of a reaction represents many things:

- ΔG_{rxn}
- Driving force
- Predictor of spontaneity if $\Delta G_{\text{rxn}} < 0$

But what's with the "free" terminology?

ΔG_{rxn} also represents the *maximum* amount of energy a spontaneous reaction can use/expend to do work.

Voltaic Cells

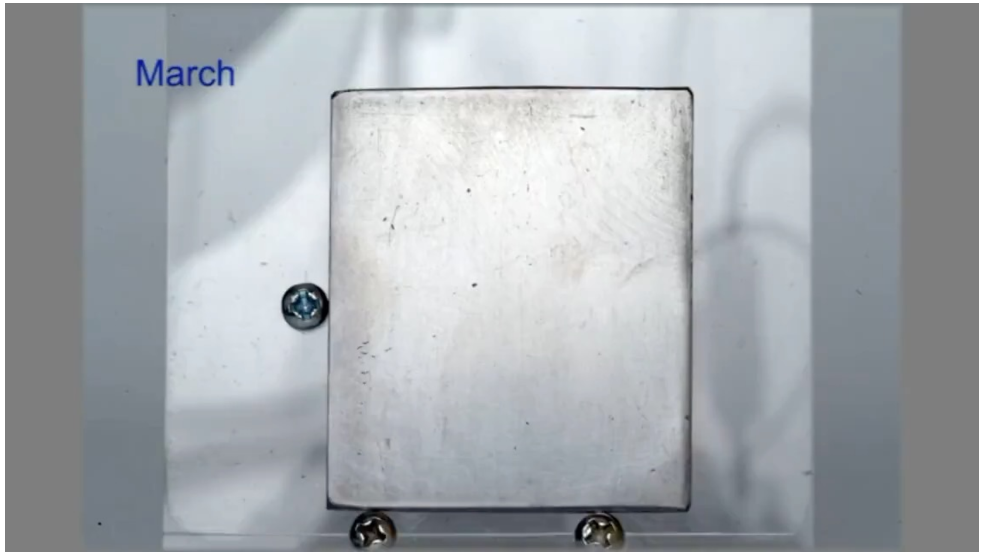
19.4

Electrochemistry

The branch of chemistry that studies the relationship between chemistry and electricity.

A *spontaneous* redox reaction is one in which $\Delta G_{\mathrm{rxn}} < 0$.

For example, rusting of iron (Fe) due to exposure to oxygen (O_2) and water (H_2O) is spontaneous.



Voltaic Cells

19.4

Electrochemical cells

A system composed of two *separated* half-cells (corresponding to two half-reactions) that are connected by a salt bridge and a wire to allow for electron transfer.

There are two types:

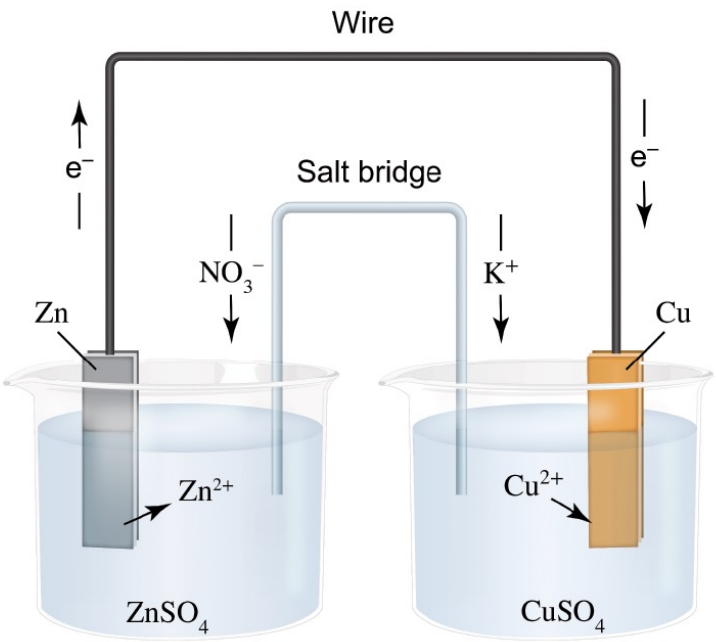
1) Voltaic cells (also called *Galvanic cells*)
    - Use *spontaneous* redox reactions to create electricity (flow of electrons).

2) Electrolytic cells
    - Use electricity to cause a *nonspontaneous* redox reaction to occur.

Voltaic Cells

19.4

Voltaic/Galvanic Cell Summary



- The overall redox reaction is *spontaneous*.
- Chemical (free) energy is converted into electrical energy.
  - Oxidation occurs at the anode half-cell.
  - Reduction occurs at the cathode half-cell.
  - Electrons flow from anode to cathode through a wire.
  - Salt bridge anions flow into the anode.
  - Salt bridge cations flow into the cathode.

⇔	red	$2\,e^-$	+	$Cu^{2+}_{(aq)}$	$\rightarrow$	$Cu_{(s)}$		
⇔	ox			$Zn_{(s)}$	$\rightarrow$	$Zn^{2+}_{(aq)}$	+	$2\,e^-$
⇔	redox	$Cu^{2+}_{(aq)}$	+	$Zn_{(s)}$	$\rightarrow$	$Cu_{(s)}$	+	$Zn^{2+}_{(aq)}$

The sustained flow of electrons produces a ____________.

The driving force for the flow of electrons is called the ___________________, measured in _________.

BONUS QUESTION

Assessed on: *undetermined*

The overall redox reaction for a lead-acid battery is:

$$PbO_2(s) + 2\ SO_4^{2-}(aq) + 4\ H^+(aq) + Pb(s) \rightarrow 2\ PbSO_4(s) + 2\ H_2O(\ell)$$

Which reactant species undergoes oxidation? *There is only one correct answer.*

- ❑ $PbO_2(s)$
- ❑ $SO_4^{2-}(aq)$
- ❑ $H^+(aq)$
- ❑ $Pb(s)$